Shafikul Islam Marwan

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github.com/Marwanthe0 | codeforces.com/profile/TheHackerBd | leetcode.com/u/marwanthe0

EDUCATION

Bachelor Of Science in Computer Science and Engineering 08/2023 – Present | Jashore, Bangladesh
Jashore University of Science and Technology
CGPA: 3.67/4.00 (till 6th semester)

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Networking, OOP

RESEARCH INTERESTS

Vector database systems and efficient vector retrieval, with a focus on approximate nearest neighbor search (ANN), HNSW and graph-based indexing, and filtered vector search on proximity graphs. Broadly interested in query optimization, adaptive indexing, CPU-oriented systems performance, and database infrastructure for modern AI/RAG workloads.
Current thesis direction: *filtered vector search on proximity graphs.*
Coursework foundation: *CMU 15-445/645 (Database Systems).*

SKILLS

- **Programming Languages:** C++, Python, C#, SQL
- **ML/Data Science:** Pandas, NumPy, Matplotlib, Seaborn, scikit-learn
- **Frameworks:** STL, FastAPI, Streamlit, Gradio, .NET, ASP.NET Core, SQL Server
- **Embedded & Hardware:** ESP32, TinyML, Arduino/C++, 1-Wire Protocol, ADC Interfacing, Relay Actuation, PWM
- **Tools:** Git, Github, VS Code, Jupyter Notebook, Google Colab, N8N, Kali Linux, Postman

ACHIEVEMENTS AND PROBLEM SOLVING

Competitive Programming: Solved **2600+ DSA problems** and participated in **100+ contests** across multiple platforms.
- **Codeforces Profiles (Current and Previous):** Max rating 1497, Problems solved 1580 (Combined)
- **Leetcode Profile:** Max rating 1739, Problems solved 186
- **Codechef Profile:** Max rating 1680, Problems solved 343
- **Vjudge Profile:** Problems solved 352
- **CSES Profile:** Problems solved 85
- **Champion** of JUST intra University Programming Contest 2025
- **National IUPC Participant** (Team: JUST BitWarriors): PSTU South Zone IUPC 2026, IUT IUPC 2026, DUET IUPC 2026.

PROJECTS

Hotel Management System | ASP.NET Core (.NET 10), React 19, SQL Server, EF Core
- Built a full-stack hotel management system with real-time analytics, booking lifecycle management, role-based access, and payment tracking using Clean Architecture.
- Implemented multi-stage booking flow with JWT authentication, BCrypt hashing, and role-based access control, validated with xUnit tests.

Github Repository

Insurance Premium Predictor | Python, FastAPI, Scikit-learn, Pandas, Streamlit
- Built a full-stack ML web app predicting health and car insurance categories using Random Forest models with feature engineering and preprocessing pipelines.
- Deployed with a FastAPI inference backend and Streamlit frontend for real-time predictions.

Github Repository , Live Demo

Smart Aquaculture: Edge-AI & Autonomous IoT Pond System | C++, ESP32, TinyML, Python, Next.js
- Designed an autonomous pond monitoring node on ESP32 interfacing pH, turbidity, temperature, and rain sensors with on-device DSP filtering for noise suppression.
- Engineered a TinyML pipeline compiling a scikit-learn decision tree into zero-heap inline C++ firmware (<1 ms inference, <1 KB flash footprint).
- Programmed hardware-level fail-safe routines, automated feeding via PWM servo control, and telemetry transmission to a full-stack dashboard.

Github Repository

ADDITIONAL SKILLS

Languages: English (Fluent), Bengali (Native), Hindi (Conversational)